Integer Programming

Name: _____ Student ID: _____

Example 7-1 Store and Storage Expansion

Hankook Supermarket has secured KRW 15.5 billion to expand by opening stores and storage facilities. Each new store costs KRW 1 billion and earns KRW 30 million per month. Each storage facility costs KRW 3 billion and earns KRW 72 million per month. Management plans to open at least one storage facility and no more than five stores.

(a) Formulate an Integer Programming model to maximize monthly profit within the available budget.
(b) Solve the model by hand using branch-and-bound, without a computer. Show the branching tree, the LP relaxation solution and upper bound at each feasible node, and the reason for pruning each closed node. Report the optimal numbers of stores and storage facilities and the maximum monthly profit.

Answer IP model

Decision variables

Objective function

Constraints and variable domains

Example 7-1 Branch and Bound Solution

Draw the tree and record each feasible node's LP solution and upper bound. Explain why each closed node is pruned.

Branching tree and node calculations

Additional calculations and pruning explanations

Final answer

Optimal decision: _____

Maximum monthly profit (include units): _____

Example 7-2 Investment Selection

H Group is considering three projects over the next three years. The table gives each project's annual funding needs and present value of profit.

Table 7.1 Investment plan (KRW 100 million)

Project	Profit PV	Year 1	Year 2	Year 3
Factory expansion	450	100	100	100
New warehouse	200	150	50	50
Advanced machinery	350	50	200	100
Available funds	—	250	400	200

Formulate a binary Integer Programming model to select projects that maximize the present value of profit, subject to the available funds in each year. Each project is either selected in full or not selected.

Answer IP model

Decision variables

Objective function

Constraints and variable domains

Example 7-3 Machine Selection and Production

K Manufacturing must produce 20,000 units using three machines. Each machine incurs a fixed cost when activated. The table lists each machine's fixed cost, variable cost, and production capacity.

Machine	Fixed cost (KRW 10,000)	Variable cost (KRW 10,000/unit)	Capacity (units)
1	300	2	8,000
2	100	10	13,000
3	200	5	14,000

Formulate an Integer Programming model to choose which machines to use and how much each should produce to minimize total production cost.

Answer IP model

Decision variables

Objective function

Constraints and variable domains